

SM-3: K3 Spectral Theorem and the Koide Formula

600-Cell Standard Model Emergence Series

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Thomas Lee Abshier, ND Grok (xAI) Claude Sonnet (Anthropic)
Claude Opus (Anthropic) ChatGPT (OpenAI)

Hyperphysics Institute
<https://hyperphysics.com>
drthomas007@protonmail.com

Abstract

We prove that the Koide relation $K \equiv \Sigma m_i / (\Sigma \sqrt{m_i})^2 = 2/3$ for the three charged lepton masses follows from the adjacency spectrum of the colour cage base graph K_3 (the equilateral triangle $\{V_1, V_2, V_3\}$ of the tetrahedral cage), conditional on three propositions whose epistemic status is made explicit via a Layer A/B/C decomposition. (P1) The ZBW Hamiltonian $\hat{H} = \hbar\omega_0 A_{K_3}$ is derived from C3 cage symmetry and SSV hopping (Layer A). (P2) $m_i \propto |\psi_i|^2$ is derived from the CPP DI-bit visit rate (Layer A). (P3) Equal eigenstate occupation follows from thermal equilibration with the DP Sea in the high-temperature limit ($kT_P/\hbar\omega_0 \sim 10^{20}$); the Gibbs formalism is standard, but the system-bath coupling model is imported from open quantum systems theory (Layer B). Under P1–P3, the eigenvalue ratio $\lambda_{\max}/|\lambda_{\min}| = 2$ forces the Lorentz modulation depth $\rho = \sqrt{2}$, from which $K = (1 + \rho^2/2)/3 = 2/3$ follows exactly (Layer C). At finite temperature, departures from $K = 2/3$ are of order $\hbar\omega_0/kT_P \sim 10^{-20}$, nine orders of magnitude below the 11 ppm experimental precision. The result is specific to K_3 : only the three-colour cage ($N = 3$) gives $K = 2/3$. The theorem applies to *leptons only*; quarks carry additional strong-sector mass contributions that systematically break Koide (deviations of 10–27% observed). The Koide phase θ is not determined by this theorem; its derivation requires the electroweak sector (OP-SM-7d).

keywords: Koide formula, lepton mass ratio, K3 spectral theorem, 600-cell lattice, Conscious Point Physics, Zitterbewegung, charged lepton masses, Standard Model emergence, parameter-free derivation, cage base graph

1 Background

The Koide formula (Koide, 1982 [6]) is an empirical constraint satisfied by the three charged lepton masses to 11 ppm:

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (1)$$

No Standard Model derivation exists. This paper proves $K = 2/3$ from the spectral structure of the K_3 cage base graph in CPP, conditional on an open-system thermalisation model whose epistemic status is made explicit via a Layer A/B/C decomposition (§3).

Note on independence: The K_3 graph is not chosen to reproduce the Koide relation. It is identified as the colour cage base of the 600-cell tetrahedral cage in SM-1 [1] from geometric considerations alone — specifically, the three colour vertices $\{V_1, V_2, V_3\}$ of the tetrahedral quark cage, whose C3 symmetry is forced by the 600-cell lattice geometry. The Koide relation emerges as a consequence of that independently-motivated structure, not as a motivation for it.

Prerequisites used without proof here:

- **Theorem 1** (Charge quantisation, SM-1 [1]): the cage base $\{V_1, V_2, V_3\}$ has C3 symmetry; $\delta = 1/3$ (topological proof from completeness + C3).
- **Algebraic identity:** with the parametrisation $\sqrt{m_i} = A(1 + \rho \cos \phi_i)$ and C3 phases $\phi_i = \theta + 2\pi i/3$, one has $K = (1 + \rho^2/2)/3$. Hence $K = 2/3 \Leftrightarrow \rho = \sqrt{2}$.
- **sea_strength** ≈ 0.1780 derived from the 600-cell Voronoi geometry (SS-1, Theorem 6 [3]).

2 The Colour Cage Base Graph

Definition 2.1 (Colour cage base graph K_3). *The three colour vertices $\{V_1, V_2, V_3\}$ of the tetrahedral cage, with edges connecting every pair, form the complete graph K_3 (equilateral triangle). Its adjacency matrix is:*

$$A_{K_3} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Lemma 2.1 (K_3 adjacency spectrum). *The eigenvalues of A_{K_3} are $\lambda_{\max} = 2$ (multiplicity 1, bonding) and $\lambda_{\min} = -1$ (multiplicity 2, antibonding). The bonding eigenvector is $(1, 1, 1)/\sqrt{3}$.*

Proof. $A_{K_3}(1, 1, 1)^T = 2(1, 1, 1)^T$. For $v \perp (1, 1, 1)$: $(A_{K_3}v)_i = \sum_{j \neq i} v_j = -v_i$, so $A_{K_3}v = -v$. $\square$

Remark 2.1 ($K = 2/3$ is unique to $N = 3$). *For the complete graph K_N , the eigenvalues are $N - 1$ (once) and -1 ($N - 1$ times). Applying the same equipartition argument gives $\rho^2 = N - 1$ and*

$$K = \frac{N + 1}{2N}.$$

Only $N = 3$ gives $K = 2/3$. The three-colour structure of the cage base is the precise reason the lepton Koide ratio is $2/3$.

3 Epistemic Layer Structure

Following the programme-wide convention established in SS-3 v1.4, this paper makes the epistemic status of its assumptions explicit by separating the argument into three layers, clearly distinguishing what comes from CPP geometry, what is imported from standard open-system quantum mechanics, and what follows mathematically.

3.1 Layer A — Geometric input (CPP primitives)

The following inputs are derived from CPP axioms:

(A1) The 600-cell lattice is composed exclusively of tetrahedral cells (Axiom A2).

- (A2) Each tetrahedral cell has 3 colour base vertices $\{V_1, V_2, V_3\}$ with C3 symmetry (SM-1 Theorem 1).
- (A3) The adjacency matrix A_{K_3} has eigenvalues $+2$ (once) and -1 (twice) — a geometric fact of K_3 (Lemma 2.1).
- (A4) The DP Sea exists at temperature T_P (Axiom A1, A3).
- (A5) The DI-bit exchange mechanism couples the ZBW orbital to the DP Sea at each Absolute Moment (Axiom A3).

These are geometric and dynamical facts of the CPP lattice. They are not postulates of this paper.

3.2 Layer B — Open-system formalism (imported structure)

The following assumptions are adopted from standard open quantum systems theory. They are *not* derived from CPP primitives in this paper:

- (B1) **Caldeira–Leggett coupling.** The ZBW–DP Sea interaction is modelled as a system-bath coupling of Caldeira–Leggett form [5]: $\hat{H}_{\text{coup}} = \sum_i \sum_k g_k |i\rangle \langle i| (b_k + b_k^\dagger)$. This is a modelling choice consistent with DI-bit exchange but not uniquely determined by it.
- (B2) **Rapid thermalisation.** The relaxation time $\tau_{\text{relax}} \ll \tau_{\text{ZBW}}$, ensuring equilibration within each ZBW cycle. No dynamical estimate of τ_{relax} is given in this paper.
- (B3) **Full Gibbs equilibration.** The system-bath coupling produces the canonical Gibbs state $\rho_{\text{eq}} = e^{-\hat{H}/kT}/Z$, not merely dephasing (decoherence without energy redistribution). This requires the bath spectral density to couple transitions between eigenstates of $\hat{H}_{\text{ZBW}}$, which is stronger than diagonal coupling alone would guarantee [4].

Remark 3.1 (Status of Layer B). *Assumptions B1–B3 are standard in open quantum systems theory and are physically motivated by the CPP DP Sea mechanism: every CP exchanges DI-bits with the DP Sea at each Absolute Moment (Axiom A3), providing a natural system-bath coupling. However, a rigorous derivation showing that DI-bit exchange dynamics forces Caldeira–Leggett coupling, rapid thermalisation, and full Gibbs equilibration has not been given. Such a derivation — deriving the operator formalism and system-bath structure from CPP primitives — would convert the present conditional result into an unconditional one. This is the same Layer B gap identified in SS-3 (Remark 3.1) and constitutes the programme’s central open problem, targeted by a future paper (SS-4).*

3.3 Layer C — Mathematical result

Given Layers A and B:

The three K_3 eigenstates are equally occupied in the high-temperature Gibbs limit. The eigenvalue ratio $\lambda_+/|\lambda_-| = 2$ then forces $\rho = \sqrt{2}$, from which $K = 2/3$ follows exactly.

This is proved in §5 (Theorem 5.1).

The layered structure makes the logical dependency transparent. Layer C is mathematically certain. Layer A is geometrically certain (given the 600-cell axiom). The open question is whether Layer B can be derived from Layer A, which would close the gap between *conditional* and *unconditional* derivation of the Koide formula.

4 The Three Propositions

Proposition 4.1 (ZBW Hamiltonian — derived from C3 symmetry and SSV hopping). *The ZBW orbital Hamiltonian restricted to the cage base is:*

$$\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3},$$

where $\omega_0 \equiv \text{sea_strength} \times c/r_{\text{conf}}$. This follows from C3 cage symmetry (Theorem 1), equal K_3 edge lengths $a = 1/\varphi$, and the SSV hopping amplitude (Proposition 4).

Remark 4.1 (Adjacency matrix vs. Laplacian). *The graph Laplacian $L_{K_3} = 2I - A_{K_3}$ has eigenvalues 0 (once) and 3 (twice). The Koide-relevant ratio $\lambda_{\text{max}}/|\lambda_{\text{min}}|$ is the same under either the adjacency matrix or the Laplacian; the adjacency matrix is the natural Hamiltonian for a nearest-neighbour hopping system, where off-diagonal elements represent hopping amplitudes between connected vertices. The result $K = 2/3$ is not an artefact of this choice.*

Proposition 4.2 (ZBW Born Rule — derived from P1 and P3). *The mass contribution of generation i is proportional to the squared ZBW wavefunction amplitude at colour vertex V_i : $m_i \propto |\psi_i|^2$. This is derived from P1 and P3 via the CPP DI-bit flow rate (Proposition 4).*

Proposition 4.3 (Thermal equipartition — from DP Sea thermalisation model (Layer B)). *At thermal equilibrium with the DP Sea, all K_3 eigenstates are equally occupied, giving $|c_+|^2 = 1/3$ and $|c_-|^2 = 2/3$. Conditional on the Layer B thermalisation model (Proposition 4; see §3.2).*

Remark 4.2 (Physical motivation for P3). *In the high-temperature limit, each quantum state has equal occupation probability $1/g$ where g is the total number of states. For K_3 : $g = 3$ eigenstates, so each has probability $1/3$. The bonding sector contributes $1/3$; the two-dimensional antibonding sector contributes $2/3$. P3 is state-counting equipartition (equal weight per eigenstate), not energy equipartition.*

Remark 4.3 (Physical motivation for P1 and P2). *P1: The ZBW orbital hops between colour vertices via the electrostatic SSV interaction. For leptons (no colour charge), the amplitude is set by the DP Sea potential $V(r) = -\text{sea_strength} \times \hbar c/r$ at the confinement radius.*

P2: Each CP processes DI-bit flows at rate proportional to $|\psi_i|^2$ (the ZBW occupation probability at vertex i). Mass = stored ZBW energy $\propto$ DI-bit flow rate. P2 is the lepton-sector analogue of the Born rule (OP-QM-1).

Derivation of the ZBW Hamiltonian from C3 Symmetry and SSV Hopping

Proposition 4.4 (C3 symmetry forces $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$). *Given the C3 cage symmetry (Theorem 1) and equal K_3 edge lengths $a = 1/\varphi$, the ZBW Hamiltonian takes the form $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$ with $\hbar\omega_0 = \text{sea_strength} \times \hbar c/r_{\text{conf}}$.*

Proof. Part A — Uniformity (C3 symmetry). The rotation $\rho : V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_1$ is an exact isometry (Theorem 1). Any Hamiltonian commuting with ρ has equal off-diagonal elements $t_{12} = t_{23} = t_{31} \equiv t$ and equal on-site energies. Setting the energy origin to zero gives:

$$\hat{H}_{\text{ZBW}} = t A_{K_3}. \quad (2)$$

Part B — Amplitude $t = \hbar\omega_0$ (SSV hopping). For leptons, the hopping amplitude is set by the electrostatic SSV potential at the confinement radius:

$$t = \frac{\text{sea_strength} \times \hbar c}{r_{\text{conf}}} \equiv \hbar\omega_0 \approx 87.8 \text{ MeV}. \quad (3)$$

Conclusion. $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$. □ □

Remark 4.4 (Consistency with SS-1 Theorem 6). *The hopping energy $\hbar\omega_0 = \text{sea_strength} \times \hbar c / r_{\text{conf}}$ is consistent with the DP binding energy $E_{\text{eDP}} = \hbar\omega_0 / \varphi^2$ from SS-1 Theorem 6 [3]. The $1/\varphi^2$ factor arises from the Voronoi volume per vertex; $\hbar\omega_0$ is the total SSV energy at r_{conf} before the per-vertex projection.*

Derivation of the ZBW Born Rule from P1 and P3

Proposition 4.5 (DI-bit flow rate gives $m_i \propto |\psi_i|^2$). *Given P1 ($\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K3}$) and P3 (thermal equipartition), the lepton mass $m_i \propto |\psi_i|^2$.*

Proof. The ZBW orbital visits each colour vertex with frequency $\Phi_i = \omega_{\text{ZBW}} \times f_i$, where f_i is the fraction of Absolute Moments spent at V_i . The mass energy stored at vertex i per unit time is:

$$m_i c^2 = \hbar \Phi_i = \hbar \omega_{\text{ZBW}} f_i. \quad (4)$$

By P1, the lepton generation state for generation g is:

$$|\psi_g\rangle = c_+ |\phi_+\rangle + c_- e^{i \cdot 2\pi g/3} |\phi_-^{(g)}\rangle, \quad (5)$$

and the occupation at vertex i is $f_i = |\langle i | \psi_g \rangle|^2 = |\psi_i|^2$. Substituting: $m_i \propto |\psi_i|^2$. $\square$ $\square$

Remark 4.5 (P3 does double duty). *P3 simultaneously fixes $|c_-|^2/|c_+|^2 = 2$ (giving $\rho = \sqrt{2}$ and $K = 2/3$) and specifies the coherent generation state, making the ZBW trajectory ergodic and giving $f_i = |\psi_i|^2$. The theorem rests on **two** independent propositions (P1 and P3).*

Derivation of Thermal Equipartition from DP Sea Thermalisation

Proposition 4.6 (DP Sea thermalisation gives $|c_n|^2 = 1/3$). *The ZBW $K3$ resonator couples to the DP Sea thermal bath at the Planck temperature T_P . Since $kT_P/\hbar\omega_0 \approx 10^{20} \gg 1$, the Gibbs state reduces to the uniform mixture $|c_n|^2 = 1/3$ for each of the three $K3$ eigenstates, giving $|c_-|^2/|c_+|^2 = 2$.*

Proof. Step 1 (ZBW–DP Sea coupling — Layer B1). Every CP exchanges DI-bits with the DP Sea at each Absolute Moment (Layer A5). This exchange is modelled as a Caldeira–Leggett system-bath coupling [5] (Layer B1, §3.2):

$$\hat{H}_{\text{coup}} = \sum_i \sum_k g_k |i\rangle \langle i| (b_k + b_k^\dagger). \quad (6)$$

This coupling form is physically motivated by DI-bit exchange but is not uniquely determined by it; see Remark 3.1 for its epistemic status.

Step 2 (Thermalisation — Layer B2, B3). We assume rapid thermalisation: $\tau_{\text{relax}} \ll \tau_{\text{ZBW}}$ (Layer B2), yielding the canonical Gibbs state (Layer B3):

$$\rho_{\text{eq}} = \frac{e^{-\hat{H}_{\text{ZBW}}/kT_P}}{Z}. \quad (7)$$

Full Gibbs equilibration (as opposed to dephasing alone) requires the bath spectral density to couple transitions between energy eigenstates, a condition that is assumed here but not derived from CPP dynamics [4].

Step 3 (High-temperature limit — Layer C).

$$\frac{kT_P}{\hbar\omega_0} \approx 10^{20} \gg 1 \implies |c_n|^2 \rightarrow \frac{1}{3} \quad \text{for each eigenstate.}$$

Step 4 (Counting — Layer C). K_3 has one bonding state and two antibonding states:

$$|c_+|^2 = \frac{1}{3}, \quad |c_-|^2 = \frac{2}{3}, \quad \frac{|c_-|^2}{|c_+|^2} = 2. \quad \square$$

□

Remark 4.6 (Finite-temperature robustness). *At finite temperature T , the exact Gibbs weights give the antibonding-to-bonding occupation ratio*

$$\frac{|c_-|^2}{|c_+|^2} = 2e^{3x}, \quad x \equiv \frac{\hbar\omega_0}{kT}. \quad (8)$$

This follows from the energies $E_+ = 2\hbar\omega_0$ and $E_- = -\hbar\omega_0$, giving Gibbs weights $e^{-2\hbar\omega_0/kT}$ (bonding) and $2 \times e^{+\hbar\omega_0/kT}$ (antibonding, degeneracy 2), so the ratio is $2e^{3\hbar\omega_0/kT}$.

For $x \ll 1$:

$$\frac{|c_-|^2}{|c_+|^2} = 2(1 + 3x + O(x^2)). \quad (9)$$

At the Planck temperature, $x = \hbar\omega_0/kT_P \sim 10^{-20}$, so the correction to the exact ratio of 2 is of order $O(10^{-20})$ — algebraically tiny, and nine orders of magnitude below the 11 ppm experimental precision of the Koide relation. Departures from $K = 2/3$ are suppressed by powers of $\hbar\omega_0/kT$, not exponentially; the robustness is algebraic, not doubly-exponential.

5 The Theorem

Theorem 5.1 (K3 spectral origin of the Koide formula). *Under Propositions 4.1–4.3 (Layers A and B, §3) and C3 cage symmetry (SM-1 Theorem 1), the three charged lepton masses satisfy $K = 2/3$ exactly.*

Proof. **Step 1 (Eigenstate structure, from P1).** The ZBW eigenstates on K_3 are the bonding mode $|\phi_+\rangle$ at energy $2\hbar\omega_0$ and two antibonding modes at energy $-\hbar\omega_0$.

Step 2 (Occupation amplitudes, from P3). Thermal equipartition: $|c_+|^2 = 1/3$, $|c_-|^2 = 2/3$.

Step 3 (Modulation depth ρ).

$$\rho^2 = \frac{|c_-|^2}{|c_+|^2} = \frac{2/3}{1/3} = 2 \implies \rho = \sqrt{2}.$$

Step 4 (Koide K, algebraic identity). With C3 symmetry (phases $\phi_i = \theta + 2\pi i/3$) and $\rho = \sqrt{2}$:

$$K = \frac{1 + \rho^2/2}{3} = \frac{1 + 1}{3} = \frac{2}{3}. \quad \square$$

□

Remark 5.1 (Physical driver). *The value $K = 2/3$ arises from the combination of the 2 : 1 eigenstate degeneracy of the K_3 adjacency spectrum and the high-temperature Gibbs weighting, which converts that degeneracy into the occupation ratio $|c_-|^2/|c_+|^2 = 2$. Both ingredients are necessary: the degeneracy is Layer A (geometry); the thermal weighting is Layer B (open-system model).*

Corollary 5.2 (Common K_3 origin of $\delta = 1/3$ and $K = 2/3$). *Both charge quantisation ($\delta = 1/3$, SM-1 Theorem 1) and the Koide formula ($K = 2/3$, Theorem 5.1) arise from the same triangle K_3 :*

Result	Value	K_3 structure used
Charge fraction δ	1/3	Combinatorial: 3 equal vertices, sum to 1
Koide ratio K	2/3	Spectral: eigenvalue ratio 2 : 1

6 Why Quarks Do Not Satisfy Koide

Remark 6.1 (Quarks: Koide broken by strong-sector mass). *The observed values are:*

$$K(d, s, b) = 0.731 \quad (9.7\% \text{ above } 2/3), \quad K(u, c, t) = 0.849 \quad (27\% \text{ above } 2/3).$$

Quarks carry qDP chain binding energy ($\sim 99\%$ of proton mass), inter-cage bonding, and cage-depth scaling. These strong-sector contributions break the K_3 spectral symmetry. Leptons carry no colour charge and no qDP chains; their masses are purely from K_3 ZBW modes, and Koide holds to 11 ppm. Theorem 5.1 is a lepton result, not a universal one.

7 Scope and Open Problems

Claim	Status	Layer / Notes
K_3 eigenvalues $\{+2, -1, -1\}$	Proved	Layer A (geometry)
C3 phase structure	Proved	Layer A (SM-1 Theorem 1)
$K = 2/3$ (Koide formula)	Proved	Layer C (conditional on A+B)
$\rho = \sqrt{2}$ from K_3 spectrum	Proved	Layer C
P1: $H = \hbar\omega_0 A_{K_3}$	Derived	Layer A (C3 + SSV hopping)
P2: $m_i \propto \psi_i ^2$	Derived	Layer A (DI-bit visit rate)
P3: Thermal equipartition	Conditional	Layer B (Gibbs model assumed; §3.2)
Robustness: $\delta K \sim O(10^{-20})$	Proved	Layer C (Remark 4.6)
Phase θ	Open	OP-SM-7d
Scale A	Calibrated to m_e	One free parameter
Individual masses m_μ, m_τ	Open	Need θ and A ; see SM-4
Caldeira–Leggett coupling	Assumed	Layer B1 (modelling choice)
$\tau_{\text{relax}} \ll \tau_{\text{ZBW}}$	Assumed	Layer B2 (no dynamical estimate)
Full Gibbs equilibration	Assumed	Layer B3 (vs. dephasing only)

Open Problem 1 (OP-SM-7d: Derivation of the Koide phase θ). *Derive $\theta = 132.7323$ from CPP dynamics. Theorem 5.1 proves $K = 2/3$ but leaves θ undetermined. A structural theorem (proved in SM-4, Theorem 2 [2]) shows that no mechanism within the $K3+SSV$ framework can select θ , because $C3$ symmetry leaves the antibonding subspace degenerate. The derivation of θ requires the electroweak sector (EW series, OP-EW-1).*

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